

bet365 NBA Points Model: Curated Extensions

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Baseline. The submitted baseline predicts the conditional mean with LightGBM Tweedie, then builds a Negative Binomial distribution using validation-binned dispersion:

$$Var(Y) = \mu + \alpha\mu^2.$$

The data are split chronologically: training fits model parameters, validation handles early stopping, calibration and model choices, and the final MAE, RMSE and CRPS values below are measured on the held-out test period.

Model	MAE	RMSE	CRPS
Original V3 LightGBM Tweedie + NB	4.37998	5.75960	3.06789
V11 direct/two-stage blend	4.36808	5.75320	3.06393
V12 hurdle + temperature CDF	4.41624	5.76313	3.06207
V13 second LightGBM variance + scale	4.37998	5.75960	3.11536

V11: direct/two-stage blend. Direct model loss: LightGBM Tweedie for points mean, early-stopped on validation RMSE. Auxiliary losses: LightGBM regression/RMSE for minutes and points per minute. The structural estimate is:

$$\widehat{points} = \widehat{minutes} \times \widehat{points/minute}.$$

Validation MAE chose:

$$\hat{\mu} = 0.65\hat{\mu}_{direct} + 0.35\hat{\mu}_{two\ stage}.$$

This is the best balanced model because it improves MAE, RMSE and CRPS.

V12: hurdle + temperature. Gate loss: LightGBM binary log-loss for $P(Y > 0)$. Positive mean loss: LightGBM Tweedie trained only on $Y > 0$. Distribution: positive-truncated NB. Calibration: one Platt/temperature-style CDF correction fitted on validation CDF events. This gives the best CRPS, showing that zero outcomes deserve special treatment, but it worsens MAE because the hurdle changes the implied mean.

V13: second LightGBM variance. Mean loss: LightGBM Tweedie. Variance loss: LightGBM regression/RMSE on $\log(1 + (Y - \hat{\mu})^2)$, using chronological out-of-fold residuals on the train period. Variance is floored at $Var \geq \mu$, then tested raw and with validation scale 3.00. It worsened CRPS, so the simple validation-binned NB layer is more robust.

Conclusion. V11 is the practical recommendation because it improves MAE, RMSE and CRPS relative to the original V3 baseline. V12 is the strongest pure distributional extension because it achieves the lowest CRPS, but its worse MAE makes it less attractive as the main replacement. V13 is a tested variance-modelling extension that is rejected because it worsens CRPS.